📐 SSC CGL — Geometry & Mensuration Master Sheet

Every figure & theorem, with recognition cues (**when to use**), scenarios, and formulas — in one tabular reference.

AIR-10 Master Repository · Quick-Recognition Edition

1 · Plane Figures (2D — Area & Perimeter)

Figure	When to use / Scenario	Area	Perimeter / Other
Square	"side", equal sides, square tiles/plot	a^2 or $d^2/2$	Perimeter $4a$, Diagonal $a\sqrt{2}$
Rectangle	"length & breadth", floor, field, room	$l \times b$	Perimeter $2(l+b)$, Diag $\sqrt{(l^2+b^2)}$
Parallelogram	slanted sides, base & height given	$base \times height$	Perimeter $2(a+b)$
Rhombus	all sides equal + diagonals given	$\frac{1}{2} \times d_1 \times d_2$	Side $\frac{1}{2}\sqrt{(d_1^2+d_2^2)}$, Per $4a$
Trapezium	two parallel sides (a, b) + height	$\frac{1}{2}(a+b) \times h$	a, b = parallel sides
Triangle (general)	base & height known	$\frac{1}{2} \times b \times h$	Perimeter = a+b+c
Equilateral △	all 3 sides equal	$(\sqrt{3}/4)\,a^2$	Height $(\sqrt{3}/2)\,a$
Regular Hexagon	6 equal sides (honeycomb)	$(3\sqrt{3}/2)\,a^2$	= 6 equilateral triangles
Regular Polygon	n equal sides	—	Interior ∠ $(n-2)180/n$, Sum $(n-2)180$

Heron's Formula — use when all 3 sides (a,b,c) are known but no height: $s = (a+b+c)/2$ → Area = $\sqrt{(s(s-a)(s-b)(s-c))}$

2 · Solid Figures (3D — Volume & Surface Area)

Solid	When to use / Scenario	Volume	Curved / Lateral SA	Total SA
Cube	"edge a", dice, box equal sides	a^3	$4a^2$	$6a^2$
Cuboid (Parallelepiped)	l, b, h given; brick, tank, room volume	$l \cdot b \cdot h$	$2h(l+b)$	$2(lb+bh+hl)$
Cylinder	pipe, tank, coin, circular base	$\pi r^2 h$	$2\pi rh$	$2\pi r(r+h)$
Cone	ice-cream, heap, tent; slant l	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(r+l)$
Sphere	ball, globe, full round	$(4/3)\pi r^3$	—	$4\pi r^2$
Hemisphere	bowl, dome, half ball	$(2/3)\pi r^3$	$2\pi r^2$	$3\pi r^2$
Prism	uniform cross-section throughout	(Base area)×h	(Perimeter)×h	Lateral + 2·Base
Pyramid	apex over polygon base	⅓(Base area)×h	½·Per·(slant)	Lateral + Base

Cone slant height: $l = \sqrt{r^2 + h^2}$ | **Frustum volume:** $\frac{1}{3}\pi h(R^2 + r^2 + Rr)$

3 · Triangle Centres, Sides & Trig

Concept	When to use / Scenario	Formula
Pythagoras	right-angled triangle, hypotenuse	$hyp^2 = base^2 + perp^2$
Triplets	spot right triangles fast	3-4-5 · 5-12-13 · 8-15-17 · 7-24-25 · 9-40-41
Inradius (r)	circle inside triangle	$r = Area / s$ (s = semi-perimeter)
Circumradius (R)	circle through all vertices	$R = abc / 4 \cdot Area$
Median (length)	vertex → midpoint opposite	$m_a = \frac{1}{2}\sqrt{(2b^2+2c^2-a^2)}$
Centroid	intersection of medians	Divides median $2 : 1$ | G = $((x_1+x_2+x_3)/3, \ldots)$
Angle Bisector	bisector splits opposite side	$BD/DC = AB/AC$
Sine Rule	angles + opposite sides	$a/\sin A = b/\sin B = c/\sin C = 2R$
Cosine Rule	2 sides + included angle	$c^2 = a^2+b^2-2ab \cdot \cos C$
Area (trig)	2 sides + included angle	$\frac{1}{2} \cdot ab \cdot \sin C$

4 · Circle — Sector, Segment, Arc, Tangent

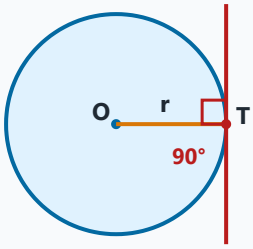
Part	When to use / Scenario	Formula
Area	radius/diameter given	πr^2
Circumference	boundary length, wheel rotations	$2\pi r$ = πd
Arc length	central angle θ given	$(\theta/360) \times 2\pi r$
Sector area	"angle at centre", pizza slice	$(\theta/360) \times \pi r^2$
Triangle (θ=90°)	chord makes right angle at centre	$\frac{1}{2}\ r^2$
Segment	chord + sector + triangle	$Sector - Triangle$
Chord length	angle θ subtended at centre	$2r \cdot \sin(\theta/2)$
Tangent (external pt)	tangent from outside point	$\sqrt{(d^2 - r^2)}$

5 · Circle & Geometry Theorems (Recognition → Rule)

Theorem	When you see this…	Apply this rule
Angle at centre	arc subtends ∠ at centre & circumference	Centre angle = `2 ×` circumference angle
Angle in semicircle	diameter is one side of triangle	Angle in semicircle = `90°`
Same segment	angles on same arc/chord	Angles in same segment are `equal`
Cyclic quadrilateral	4 points on a circle	Opposite angles sum `= 180°`
Tangent ⊥ radius	tangent touches circle	Tangent ⊥ radius at contact (`90°`)
Two tangents	tangents from one external point	Equal length: `PA = PB`
Alternate segment	tangent + chord angle	= angle in alternate segment
Intersecting chords	two chords cross inside	$PA \cdot PB = PC \cdot PD$
⊥ chords → radius	two chords cross at 90°; distances d_1, d_2 from centre	$R = \sqrt{d_1^2 + d_2^2}$ (e.g. 5,12 → 13)
Secant–Tangent	tangent + secant from outside	$PT^2 = PA \cdot PB$
Equal chords	chords of equal length	Equidistant from centre
Perp. from centre	perpendicular drops to chord	Bisects the chord
BPT (Thales)	line parallel to one side of △	$AD/DB = AE/EC$
Similar triangles	AA / equal angles / parallel	Sides proportional; Area ratio = $(\text{side ratio})^2$
Apollonius	median to a side	$AB^2+AC^2 = 2(AD^2+BD^2)$
Ptolemy	cyclic quadrilateral diagonals	$AC \cdot BD = AB \cdot CD + AD \cdot BC$

6 · 📊 Tangent & Circle Theorems — Visual Diagrams

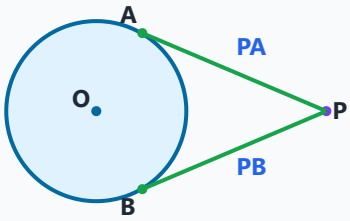
① Tangent ⊥ Radius



Radius meets the tangent at the point of contact T.

$OT \perp \text{tangent} \rightarrow \text{angle} = 90°$

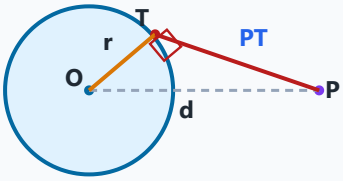
② Two Tangents from a Point



Two tangents drawn from one external point P.

$PA = PB$ (equal lengths)

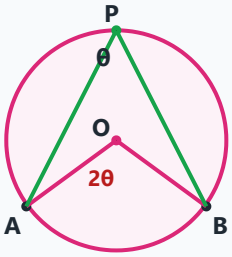
③ Tangent Length (PT)



Length of tangent from external point, distance d = OP.

$PT = \sqrt{(d^2 - r^2)}$

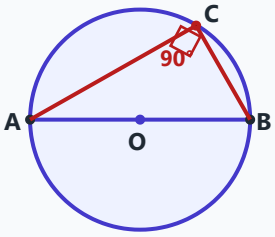
④ Angle at Centre = 2 × Circumference



Arc AB subtends ∠ at centre (O) and on circumference (P).

$\angle AOB = 2 \times \angle APB$

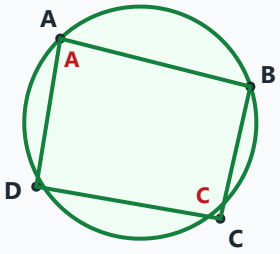
⑤ Angle in Semicircle = 90°



AB is a diameter; C is any point on the circle.

$\angle ACB = 90°$

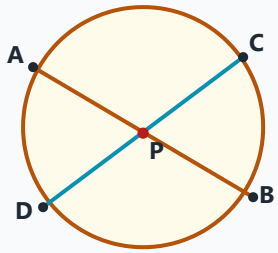
⑥ Cyclic Quadrilateral



All four vertices A, B, C, D lie on the circle.

$\angle A + \angle C = 180°$ and $\angle B + \angle D = 180°$

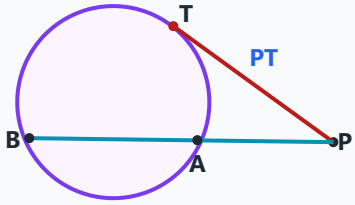
⑦ Intersecting Chords



Two chords AB and CD cross inside the circle at P.

PA × PB = PC × PD

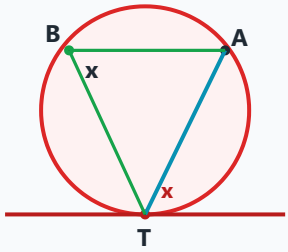
⑧ Tangent–Secant (Power of Point)



From external P: one tangent PT and one secant cutting at A, B.

$PT^2 = PA \times PB$

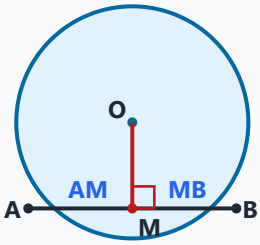
⑨ Alternate Segment Theorem



Angle between tangent & chord TA = angle in the alternate segment (∠B).

∠(tangent, chord) = ∠ in alternate segment

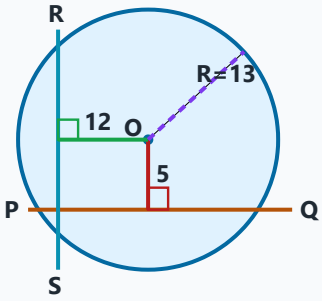
⑩ Perpendicular from Centre Bisects Chord



Perpendicular OM dropped from centre O to chord AB.

AM = MB (M is midpoint)

⑪ Two ⊥ Chords → Radius (5-12-13 PYQ)



Two chords cross at 90°; distances from centre are d_1, d_2.

$R = \sqrt{(d_1^2 + d_2^2)} = \sqrt{(5^2+12^2)} = 13$ cm

7 · Coordinate Geometry

Concept	When to use	Formula
Distance	length between 2 points	$\sqrt{((x_2-x_1)^2+(y_2-y_1)^2)}$
Midpoint	centre of a segment	$((x_1+x_2)/2,\ (y_1+y_2)/2)$
Section (m:n)	point dividing segment in ratio	$((mx_2+nx_1)/(m+n),\ \ldots)$
Centroid	3 vertices of triangle	$((x_1+x_2+x_3)/3,\ (y_1+y_2+y_3)/3)$
Area of △	3 coordinate vertices	$\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$
Slope	steepness / parallel-perp check	$m = (y_2-y_1)/(x_2-x_1)$
Parallel / Perp	relation between 2 lines	Parallel $m_1=m_2$ · Perp $m_1 \cdot m_2=-1$
Line equation	slope + intercept	$y = mx + c$

8 · ⭐ 10-Second Recognition Card

When you read / see…	Immediately think…
"In Arithmetic Progression" / 3 terms	`a−d , a , a+d` · middle = Sum/3
Dimensions + Volume	`l × b × h`
Chord + centre angle	`Segment = Sector − Triangle`
Angle θ at centre	`Sector = (θ/360)πr²`
Right angle in triangle	Pythagoras / triplet / `½·base·perp`
All 3 sides known, no height	Heron `√(s(s−a)(s−b)(s−c))`
Diameter is a side of triangle	Angle = `90°` (semicircle)
4 points on a circle	Opposite ∠ sum `180°` (cyclic)
Tangent from external point	`√(d²−r²)` · two tangents equal
Two chords cross at 90° (distances d_1, d_2)	`R = √(d₁²+d₂²)` · spot 5-12-13
Line ∥ one side of triangle	BPT `AD/DB = AE/EC`
"Average"	`Average = Sum ÷ Number`
Pipe / tank / coin (round base)	Cylinder `πr²h`
Heap / tent / ice-cream	Cone `⅓πr²h` · slant `√(r²+h²)`